ICAMDS-SFC Proceedings, September. 14-18, 2026
ISBN : 978-9920-40-160-9

Passive Sonar Ship Classification under Vessel-Disjoint Evaluation: A Benchmark Study on ONC Data

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Abstract
Deep learning methods for passive sonar ship classification routinely report accuracies above 95%, often under evaluation protocols that are seldom made explicit. This work investigates whether such performance reflects genuine generalization to vessels not seen during training, or rather the memorization of individual ship acoustic signatures present in both training and test sets. The task throughout is vessel-*class* classification; the question is whether a class is learned in a way that transfers to new vessel instances of that class, or only to the specific ships seen in training. Using the Ocean Networks Canada (ONC) hydrophone dataset, enriched with AIS metadata providing unique vessel identifiers (MMSI), we compare two evaluation protocols: a naive segment-level split, standard in the literature, and a vessel-disjoint split ensuring that no vessel seen during training appears in the test set. Both protocols are applied to three models: a linear probe on pretrained audio representations, a compact convolutional network trained on log-mel spectrograms, and an end-to-end fine-tuned Audio Spectrogram Transformer serving as a strong baseline. Our results reveal a substantial performance gap $\Delta = \text{Performance(naive)} - \text{Performance(vessel-disjoint)}$, showing that the evaluation protocol is a dominant factor in reported performance. The fine-tuned transformer reaches 93.8% accuracy under the naive split, yet its macro-F1 collapses by 0.50 under vessel-disjoint evaluation, falling close to chance. Identity probes, label-shuffle controls, and a same-condition analysis attribute the gap to individual-vessel identity rather than class scarcity or recording channel. These findings argue for vessel-disjoint splits as a minimal standard for future passive sonar benchmarks.

Résumé
Les méthodes d'apprentissage profond appliquées à la classification de navires par sonar passif annoncent régulièrement des précisions supérieures à 95%, souvent sous des protocoles d'évaluation rarement explicités. Ce travail examine si ces performances reflètent une véritable généralisation à des navires non vus à l'entraînement, ou plutôt la mémorisation de signatures acoustiques individuelles présentes à la fois en entraînement et en test. En utilisant le jeu de données hydrophones d'Ocean Networks Canada (ONC), enrichi de métadonnées AIS fournissant des identifiants uniques de navires (MMSI), nous comparons deux protocoles d'évaluation : un découpage naïf au niveau des segments, standard dans la littérature, et un découpage vessel-disjoint garantissant qu'aucun navire vu à l'entraînement n'apparaît en test. Nos résultats révèlent un écart de performance substantiel, montrant que le protocole d'évaluation est un facteur déterminant des performances rapportées. Des sondes d'identité, des contrôles par permutation d'étiquettes et une analyse à conditions constantes attribuent cet écart à l'identité individuelle des navires. Ces observations plaident pour l'adoption du découpage vessel-disjoint comme standard minimal dans les futurs benchmarks de sonar passif.

Keywords
Passive sonar, Ship classification, Evaluation protocol, Data leakage, Vessel-disjoint split, Deep learning, Ocean Networks Canada

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Introduction

Passive sonar classification is a longstanding problem in underwater acoustics, with applications ranging from maritime surveillance to environmental monitoring. Recent years have seen a surge of deep learning approaches to ship-radiated noise classification on benchmarks such as DeepShip [1], with reported accuracies above 95% [2]. Specific individual ships can moreover be identified from their radiated noise, even within multi-ship mixtures, with up to 99.56% accuracy [3], direct evidence that each vessel carries a highly recoverable acoustic fingerprint. However, a closer examination of the experimental setups reveals a recurring methodological ambiguity: the evaluation protocol used to partition data into training and test sets is rarely specified, and almost never accounts for vessel identity.

In most published studies, audio recordings are split at the segment level, meaning that multiple segments from the same physical vessel may appear on both sides of the train/test boundary. This creates a subtle but critical form of data leakage: the model may learn to recognize *individual ships* rather than *vessel classes*. Such leakage, where information about a test instance leaks into training through an uncontrolled grouping variable, has been identified as a leading cause of overoptimistic, irreproducible results across machine-learning-based science [4]. The resulting accuracy metrics are therefore misleading, reflecting memorization rather than generalization. A recent survey of the field observes that train/test splitting practices differ across studies and that randomly sampling windowed data introduces a bias that inflates reported accuracies, hindering fair comparison [5]; the present work measures that bias directly.

This work quantifies this effect systematically. We introduce a vessel-disjoint evaluation protocol, in which the sets of vessels used for training and testing are strictly disjoint, and compare it against the naive segment-level split on the Ocean Networks Canada (ONC) dataset. To our knowledge, this is among the first studies to explicitly benchmark these two protocols on a large-scale public passive sonar dataset with vessel-level traceability. Beyond the headline gap, we conduct four mechanism analyses (identity probes, per-class breakdowns, label-shuffle controls, and a same-condition control) that attribute the gap to individual-vessel identity and rule out the principal competing explanations.

1. Background and Related Work

1.1 Passive Sonar Classification

Ship acoustic signatures originate from propulsion systems (propellers, engines) and auxiliary equipment. Classical methods exploit frequency-domain representations such as LOFAR (Low Frequency Analysis and Recording), which isolates direct narrowband engine and machinery frequencies (0 to 10 kHz), and DEMON (Detection of Envelope Modulation on Noise), which demodulates high-frequency propeller cavitation noise to extract ultra-low modulation frequencies (0 to 50 Hz) [6]. More recent approaches replace handcrafted features with deep convolutional networks trained on spectrogram images [2, 7], achieving high accuracy on controlled benchmarks.

1.2 The Data Leakage Problem in Audio Classification

Identity- or group-level leakage is a recurring failure mode whenever multiple correlated samples share a hidden grouping variable that is not held constant across the train/test boundary. Kapoor and Narayanan [4] catalogue it as one of the most common errors behind irreproducible machine-learning results, and document cases across dozens of fields; in medical imaging, splitting scans rather than patients was shown to inflate reported COVID-19 diagnosis performance to the point of clinical uselessness [8]. The same pattern recurs in audio: speaker recognition, environmental sound classification, and medical audio analysis all require that recording or subject identity be disjoint across splits, lest a model recognize the recording rather than the target concept. Passive sonar is a textbook instance of this hazard, because each vessel has a characteristic acoustic fingerprint shaped by its specific machinery, hull, and operating conditions. A model trained on segments from vessel v and tested on other segments from the same vessel v is solving a fundamentally easier problem than one required to generalize to an entirely unseen vessel.

Formally, let V_{train} and V_{test} denote the sets of vessel identities (MMSI) in the training and test sets respectively. The naive split imposes no constraint on the overlap $V_{\text{train}} \cap V_{\text{test}}$, whereas the vessel-disjoint split enforces $V_{\text{train}} \cap V_{\text{test}} = \emptyset$.

1.3 Positioning with Respect to Concurrent Work

Concurrent work by Hummel et al. [9] analyzes frozen pretrained audio embeddings of ship-radiated noise and finds that the geometrical structure of the embedding space is largely dominated by recording-specific characteristics, a representation-level diagnosis that converges with the identity probes we report in Section 2.5. They further conclude, more

optimistically, that a linear probe can suppress this recording-specific information and isolate ship-type features. Our results qualify this conclusion: whether a probe has isolated class information rather than identity information can only be adjudicated under vessel-disjoint evaluation, and under our protocol a linear probe on frozen embeddings collapses from 0.769 to 0.328 macro-F1 once seen vessels are removed from the test set. The two studies are therefore complementary (representation-level there, protocol-level here) and their juxtaposition illustrates the central point of this paper: the same frozen features support opposite conclusions about transfer learning depending on the evaluation protocol, which is precisely why the protocol must become an explicit, reported design choice.

2. Methodology

2.1 Dataset: Ocean Networks Canada

We use hydrophone recordings from the Ocean Networks Canada (ONC) observatory, processed through the open pipeline of [10]. AIS (Automatic Identification System) metadata provides a unique MMSI identifier for each vessel, propagated to every audio segment, which makes vessel-level traceability explicit and vessel-disjoint splitting directly implementable. Identity metadata is not unique to this corpus: DeepShip [1], collected at the same observatory, was itself assembled by matching AIS tracks to single-ship passages and identifies the 265 individual ships underlying its recordings. What is missing in the field is not the metadata but its use: published evaluations almost universally split at the segment level without grouping by vessel, and the reference ONC pipeline likewise partitions data through a random segment-level sort, which corresponds to the naive protocol we examine in this work. Vessel-disjoint splitting must be implemented explicitly on top of the MMSI annotations.

Audio is segmented into one-minute normalized clips, and only intervals during which a single vessel is within range of the hydrophone are retained, which yields clean single-source labels. We consider the four vessel classes provided by the pipeline, namely tankers, cargo ships, passenger ships, and tugboats, and exclude the background (no-vessel) class. Recordings span multiple deployments and environmental conditions, with concurrent CTD measurements (temperature, salinity, and sound speed) available for each segment. The resulting corpus comprises 7 658 one-minute clips drawn from 296 distinct vessels, with a pronounced class imbalance (Table 1): tugboats dominate, while the rare classes are represented by very few vessels (19 tankers, 22 passenger ships). Because the many-clips-per-vessel structure is precisely the surface through which leakage occurs, and because accuracy alone can be inflated by the dominant class, we report accuracy alongside balanced accuracy and macro-F1.

2.2 Evaluation Protocols

We implement and compare two split strategies:

Table 1. Composition of the ONC corpus after segmentation into one-minute clips.

Class	1-min clips	Distinct vessels (MMSI)
Tanker	145	19
Cargo	1 892	149
Passenger ship	1 095	22
Tug	4 526	106
Total	7 658	296

- **Naive split:** audio segments are randomly assigned to train, validation, and test sets without any constraint on vessel identity. This is the de facto standard in the literature.
- **Vessel-disjoint split:** vessels are first partitioned by MMSI, then all segments from a given vessel are assigned exclusively to one subset. No vessel present in training appears in test.

2.3 Models

Three models are evaluated under both protocols:

- **Linear probe:** a class-balanced logistic regression classifier trained on top of frozen embeddings extracted from a pretrained Audio Spectrogram Transformer (AST). Each one-minute clip is resampled to 16 kHz, split into six 10 s windows, and the pooled window embeddings are averaged into a single 768-dimensional clip vector. This setup tests whether the pretrained representations themselves encode vessel-identity information.
- **Compact CNN:** a four-block convolutional network (Conv–BN–ReLU–MaxPool $\times 4$, global pooling, fully connected head) trained from scratch on log-mel spectrograms, following the general architecture of [7], with class-weighted cross-entropy and early stopping on validation macro-F1.
- **Fine-tuned AST:** the same pretrained AST fine-tuned end-to-end on the sonar classes, serving as a deliberately strong, literature-grade baseline. Training is performed at the 10 s window level with class-weighted cross-entropy, and the six window logits per clip are mean-pooled into a single clip-level prediction, making its metrics directly comparable to the other two models. Its role is to show that even a classifier reaching the field's $>90\%$ naive regime collapses under vessel-disjoint evaluation.

2.4 Central Metric

The primary quantity is the performance gap:

$$\Delta = \text{Performance(naive)} - \text{Performance(vessel-disjoint)} \quad (1)$$

A large Δ indicates that reported performance is strongly influenced by the evaluation protocol rather than genuine generalization. The gap is computed for accuracy, balanced accuracy, and macro-F1. Because the vessel-disjoint test set draws only

a handful of vessels for the rare classes, raw accuracy is sensitive to the particular vessels sampled; we therefore run the full protocol over five random seeds $\{42, 0, 1, 2, 3\}$ and report the gap on macro-F1 and balanced accuracy, which are immune to test-set class composition, as the primary metrics. Statistical significance of every gap is assessed with a paired t-test across seeds and a 95% bootstrap percentile confidence interval on Δ. A t-SNE (t-distributed Stochastic Neighbor Embedding) projection of learned representations, colored by vessel identity (MMSI) rather than class label, provides complementary qualitative evidence.

2.5 Mechanism Analyses

Four analyses test *why* the gap arises and rule out competing explanations. Unless noted, all operate on the cached frozen AST embeddings.

Identity probe. We train a class-balanced logistic regression to predict the *MMSI* of a clip rather than its class, restricted to the 103 vessels with at least ten clips, with every vessel present on both sides of a stratified split (chance $1/103$). A within-class variant predicts the individual vessel among those of a single class, and therefore carries no class information by construction.

Per-class breakdown. We report per-class F1 under both protocols. The vessel-rich classes (cargo, 149 vessels; tug, 106) separate leakage from scarcity.

Label-shuffle controls. Using the same splits, we replace the true labels with random ones, assigned either per clip (*clip-level*) or per vessel (*vessel-level*, so labels remain vessel-consistent but acoustically meaningless). The first is a pipeline sanity check; the second isolates identity memorization from genuine class learning.

Same-condition control. Splitting by vessel may also change the acoustic channel. We characterize this confound through the recording date and the concurrent CTD sound speed, and re-measure the gap within a narrow sound-speed band holding the environment approximately constant.

3. Results and Discussion

Table 2 reports the full multi-seed results for the three models under both protocols. The leakage effect is large and in the predicted direction for every model: each one collapses once vessels can no longer leak across the split.

The strong baseline confirms the effect at high capacity. The fine-tuned AST is the strongest of the three models, reaching 0.938 ± 0.010 accuracy (macro-F1 0.920 ± 0.012) under the naive split, squarely in the >90% regime the field reports. Yet under the vessel-disjoint protocol its macro-F1 falls to 0.420 ± 0.087 (a gap of $\Delta_{\text{macroF1}} = 0.500 \pm 0.095$) and its balanced accuracy drops to 0.480, barely above the 0.25 four-class chance level. The high naive number is therefore vessel-identity memorization rather than class generalization, and the collapse is not an artefact of weak models.

The frozen representation already encodes vessel identity. The linear probe, operating on frozen AST embeddings, exhibits a macro-F1 gap of 0.441 ± 0.111, dropping from 0.769 to 0.328. Since the features are never updated, this directly shows that the pretrained representation organizes information by individual vessel: removing seen vessels from the test set drives performance toward chance. This is the protocol-level counterpart of the recording-specific dominance reported at the representation level by [9]. The compact CNN trained from scratch shows the same pattern with a smaller but clearly positive macro-F1 gap of 0.263 ± 0.069.

Macro-F1 is the metric of record. Raw vessel-disjoint accuracy is noisy across seeds (std up to ± 0.17), because the rare classes draw only a few vessels into each test fold; the CNN's accuracy gap of 0.165 ± 0.135 is, on its own, barely distinguishable from zero. The composition-immune metrics tell a consistent and statistically clearer story: across the five seeds, every model's macro-F1 and balanced-accuracy gaps have a 95% bootstrap confidence interval that excludes zero (paired t-test $p < 0.003$ in every case), whereas the CNN's accuracy gap is the only one whose paired test is not significant ($p = 0.07$). The effect is thus statistically robust precisely on the composition-immune metrics. Strikingly, under vessel-disjoint evaluation every model falls below the majority-class baseline of 0.641 on raw accuracy (linear probe 0.483, CNN 0.593, AST 0.597): grouped evaluation leaves all three beaten by the trivial "always predict tug" rule, even though they retain weak above-chance signal (0.41–0.48 balanced accuracy versus the 0.25 four-class chance level).

Figure 1 offers qualitative support for the same reading: in a t-SNE projection of the frozen AST features colored by MMSI, several vessels occupy their own distinct regions of the embedding space, and the visible structure does not align with class boundaries. The projection is illustrative only; the quantitative identity probe below (Table 3) is the substantive evidence for this mechanism.

3.1 The mechanism is individual-vessel identity

The frozen representation encodes individual identity, quantitatively. Table 3 makes the t-SNE picture precise. A linear classifier recovers the individual vessel (MMSI) of a clip with 0.652 top-1 accuracy among 103 vessels (67× the 0.010 chance level, and 0.854 at top-5), comparable to the 0.851 four-class accuracy the same probe achieves on the same clips. The within-class probe is decisive: even restricted to a single class, where no class information is available, the representation identifies which individual cargo vessel (0.905 among 26), passenger ship (0.951 among 10), or tanker (1.000 among 4) a clip belongs to, and which tug at 0.598 among 63. The pretrained features thus organize the data by individual ship: exactly the information a segment-level split leaks across the train/test boundary.

The gap persists under meaningless labels. As a control (Section 2.5), we replace the true labels with random ones and

Table 2. Performance under naive and vessel-disjoint (VD) protocols, mean ± std over five seeds. Δ is the naive−vessel-disjoint gap. Macro-F1 and balanced accuracy are the primary metrics; raw accuracy is reported with its larger variance. Majority-class baseline: naive 0.591, vessel-disjoint 0.641 ± 0.099; four-class chance is 0.25.

	Accuracy			Balanced acc.			Macro-F1		
Model	Naive	VD	Δ	Naive	VD	Δ	Naive	VD	Δ
Linear probe	.822±.015	.483±.169	.339±.168	.781±.041	.413±.048	.368±.080	.769±.025	.328±.091	**.441±.111**
Compact CNN	.758±.051	.593±.160	.165±.135	.671±.032	.461±.055	.209±.064	.660±.040	.397±.079	**.263±.069**
Fine-tuned AST	.938±.010	.597±.149	.341±.146	.912±.020	.480±.057	.431±.058	.920±.012	.420±.087	**.500±.095**

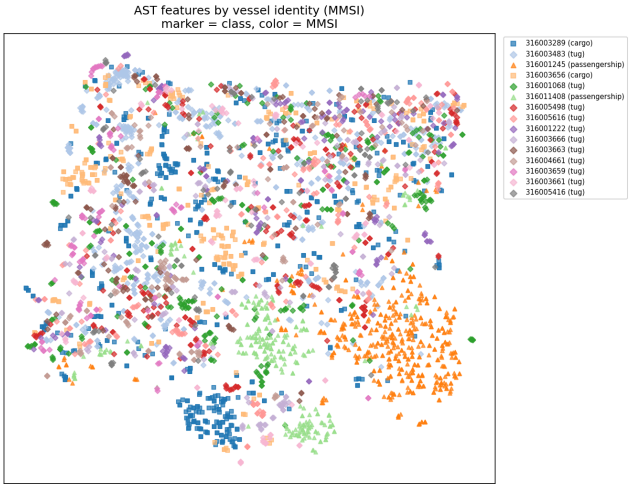


Figure 1. t-SNE of frozen AST features for the most frequent vessels (color = MMSI, marker = class). Several vessels occupy distinct regions of the embedding space, and the visible structure is not organized by class. The projection is qualitative and illustrative; see Table 3 for the quantitative identity probe.

re-run the linear probe. With clip-level random labels both protocols sit at the 0.250 chance level and the gap vanishes ($\Delta_{\text{acc}} = -0.006 \pm 0.023$), confirming that the split and evaluation machinery introduce no spurious gap. With vessel-level random labels, meaningless but consistent within each vessel, the naive split nonetheless reaches 0.564 accuracy while vessel-disjoint collapses to 0.304 ($\Delta_{\text{acc}} = 0.260 \pm 0.042$). A gap of this size under labels that carry no acoustic class meaning can only arise from memorizing identity, which attributes the naive advantage to identity leakage rather than to genuine class learning.

3.2 Leakage, not class scarcity

A natural objection is that macro-F1 is dominated by the rare classes (19 tankers, 22 passenger ships), so the collapse might reflect too few vessels to learn from rather than leakage. Table 4 refutes this. The two vessel-rich classes also collapse: cargo, with 149 distinct vessels, drops from 0.897 to 0.429 F1 ($\Delta = 0.468$), and tug, with 106, from 0.952 to 0.729 ($\Delta = 0.223$). With over a hundred vessels each, scarcity cannot explain these gaps, whereas identity leakage can. The pattern

Table 3. Identity probe on frozen AST features: closed-set top-1 accuracy of predicting individual vessel identity (MMSI). The within-class rows predict which vessel a clip belongs to among the vessels of a single class, and therefore carry no class information. Mean over five seeds (std ≤ 0.02 throughout).

Target	Vessels	Chance	Identity top-1
All vessels (global)	103	0.010	**0.652**
within Tanker	4	0.250	1.000
within Cargo	26	0.038	0.905
within Passenger	10	0.100	0.951
within Tug	63	0.016	0.598

is consistent across models: the cargo gap is 0.375 for the linear probe and 0.147 for the compact CNN, and the tug gap 0.233 and 0.114 respectively, all positive.

Table 4. Per-class F1 for the fine-tuned AST under both protocols (mean over five seeds). The vessel-rich classes cargo and tug collapse substantially, so the gap is not an artefact of having too few vessels in the rare classes.

Class	Vessels	Naive	Vessel-disjoint	Δ
Tanker	19	0.875	0.024	0.851
Cargo	149	0.897	0.429	0.468
Passenger	22	0.956	0.497	0.459
Tug	106	0.952	0.729	0.223

3.3 Disentangling identity from recording channel

Splitting by vessel can also separate recording conditions, since each vessel is recorded under a limited range of channels: 70% of vessels appear on a single date, and the recording date is itself recoverable from the frozen features at 0.602 accuracy (133 date-groups, chance 0.008). Channel leakage is therefore real, and we do not claim otherwise. It does not, however, account for the gap. First, the heavily sampled vessels that dominate the corpus are recorded across several dates: single-date vessels contribute only 17% of all clips. Second, and decisively, when we restrict the linear probe to a narrow sound-speed band (1485.6–1486.9 m/s, 1 527 clips, 100 vessels) that holds the environment approximately constant, the naive-versus-vessel-disjoint class-accuracy gap persists at 0.295 (0.881 vs 0.587), essentially its full-corpus

magnitude. A gap that survives a near-constant channel is driven by vessel identity, not by recording conditions. Sound speed summarizes only part of the environment, so this control is corroborative rather than exhaustive.

Taken together, these results show that a significant portion of the performance reported in the passive sonar classification literature is attributable to evaluation protocol choice rather than model quality. This has direct implications for how future benchmarks should be designed and how existing results should be interpreted.

3.4 Limitations and scope

Our claims are scoped to the ONC corpus. It is currently, to our knowledge, the only large public passive sonar dataset whose AIS-derived MMSI annotations make a controlled protocol comparison possible at this scale, which is precisely why it was chosen. It also means, however, that our quantitative gaps should not be transposed as-is to other corpora. Two independent lines of evidence nonetheless suggest the phenomenon is not ONC-specific: the finding of [9], obtained on different data, that recording-specific characteristics dominate the geometry of pretrained audio embeddings of ship noise, and the 99.56% accuracy with which specific individual ships are identified from their radiated noise, even in multi-ship mixtures, on QiandaoEar22 [3]: per-ship acoustic fingerprints this recoverable are exactly the signal a segment-level split leaks across the train/test boundary. Replication on additional corpora with vessel-level traceability, as MMSI-annotated datasets become available, is left to future work. Finally, the same-condition control relies on sound speed as an environmental summary and is corroborative rather than exhaustive; a fuller channel characterization would require multi-deployment recordings of the same vessels.

Conclusion

This work highlights a critical but underexplored methodological issue in passive sonar ship classification: the conflation of segment-level and vessel-level generalization in standard evaluation protocols. Our comparative study on ONC data demonstrates that the choice of split strategy substantially affects reported performance, and argues for vessel-disjoint evaluation as a minimal standard for the field. A preliminary learning-curve analysis found that the frozen AST probe did not benefit from additional training-vessel diversity, consistent with its representation organizing information by identity rather than by type; whether trainable models benefit from vessel-diverse training is the natural constructive follow-up. Future work will extend this analysis to a wider range of architectures and explore strategies to improve genuine generalization to unseen vessels, including domain adaptation and representation learning under vessel-disjoint constraints.

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